

Assessing Proximity and Access to HealthCare Services in Chicago

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Applications in GIS

Introduction:

There are many health care services in Chicago, but this paper assesses their proximity to public schools. Using spatial analysis, an assessment of health care services proximity to public schools was performed and then the density was found. These services are especially important for students to have and are essential for creating a healthy, welcoming environment where students feel safe and cared for.

There are multiple factors that influence the access and utilization of health services, but one major factor is the distance. This often acts as a barrier because it makes it harder for people who, for instance, do not have a car and cannot travel long distances. For example, a parent who works all day and only has time off in the evenings to take their students to healthcare services, will be more willing and able to take their child to a service offered closer than farther away. Additionally, many people do not own a car, which means they have to use public transportation to get to these centers. In 2015, 27.5% of Chicagoans did not own a car, and this statistic is particularly concentrated in neighborhoods with high poverty rates [1]. This is why access to public transportation and proximity of the routes to healthcare services is critical in maintaining a healthy, stable environment where people want to live. More Chicagoans do not own a car, as the wave of young people living in the city increases, and Chicago has the second largest transit system in the country. Along the northern lakefront of Chicago, almost half the households do not have cars, and some places in Near North Side and Lake View have over 60 percent of the population without cars. While the categories of one person, one vehicle and two people, one vehicle have stayed roughly the same since 2010, one person, no vehicle has increased [2]. To be able to live carless in Chicago, one must have access to everything they may need, such as grocery stores, health care, and community and transportation routes. Access to health services also includes many other factors, like cultural, behavioral, and financial factors. The distance

decay effect comes into play here, which states that as the interaction between two locales declines, the distance between them increases [3]. As the commuting distance increases, people are less likely to visit their primary health service, even if they are in need. The bus route locations were analyzed in relation to the location of health care services. The more bus route stops that are closer to each of the primary health care locations, the more likely one is to travel using the bus routes.

Before COVID-19, health disparities among Chicagoans were already severe, and COVID-19 only heightened those disparities. The poverty rate in Chicago is 20.6%, which means that one out of every 4.8 residents of Chicago live below the poverty line [4]. The number of wealthy people in Chicago has quadrupled since 1970, but people below the poverty line are still struggling, and the gap between the rich and the poor widens. Some people are also unable to travel to areas because of gang territory out of fear of being violently attacked [5]. The neighborhoods with higher unemployment rates and higher levels of poverty were examined closely because these were the places predicted to have the lowest amount of bus stops and the least amount of health care locations.

I will also be examining data on the number and location of hospitals in Chicago. I am posing the question of how well the public transportation system brings travelers to hospitals. Hospitals are in higher demand since the Coronavirus spread, and they also have some of the best hospitals in the country; Rush University Medical Center has made the list of the top 20 hospitals in the country as well as Northwestern Memorial Hospital [6]. Chicago has good healthcare when looking at this information, but we must also examine access to get the full picture.

Methods

Data

Data on the number and location of bus stops in Chicago was used to determine accessibility to health care service through public transportation. Data on the number and location of health care services were gathered and analyzed with reference to the location of public schools in the area. Hospital data was also analyzed in the same way health care service locations were, as another form of health care data. All the data on public school locations, health care locations, hospitals and bus stop locations were obtained from the City of Chicago official website. The data can be accessed through the Chicago data portal, at <https://data.cityofchicago.org/>.

Proximity of Health Care Services to Public Schools

Data on the number and location of public schools and primary health care locations were used to determine the proximity of health care services to public schools. I selected each health care service within the distances of 150, 300, 450, 600, and 750 meters to the public schools close by. For each distance, the areas were found by making buffer zones, as shown in *table 1*. The density was calculated by dividing the number of healthcare services that are within each radius of the healthcare places by the area of each region.

Proximity of Bus Stops to Health Care Services

I selected each bus stop within the distances of 150, 300, 450, 600, and 750 meters to the health care services close by. From the circles calculated, the areas were found by making buffer zones, as done for the proximity of health care services to public schools. With the circles shown on the map, the area and density were calculated as before. Each bus stop was assigned to the nearest health care location and divided into Thiessen polygons based on the health care locations. The Thiessen polygons were joined with the data on bus stops and each polygon was

displayed on a map. The number of bus stops within each Thiessen polygon was displayed, with the average number of stops, and minimum and maximum data displayed. Then, the distance of the bus stops from the healthcare place was calculated. For the Thiessen polygon data, a map was formed displaying the number of bus stops within each Thiessen polygon and sorted into 5 groups. These results were put on another map with neighborhood boundaries, to uncover any patterns between the unemployment and poverty rates within community areas, and the polygons created for the healthcare locations and the proximity of bus stops.

Proximity of Bus stops to Hospitals

Similar to the proximity of bus stops to health service locations, I selected each bus stop within the distances of 150, 300, 450, 600, and 750 meters from the hospitals close by. Buffer zone, area, and densities were calculated the same way they were in the previous section. The bus stops were joined to the Thiessen polygons created, which created one polygon for each hospital. The data showed the number of hospitals within each of the polygons, which was then sorted into different groups and displayed on a map. These results were overlayed on a map with the neighborhood boundaries to assess correlations between the location and neighborhood, and the number of bus routes and hospitals within the area.

Results

Table 1. The Proximity of Primary Health Care Services to Public Schools

Distance from Schools (m)	Number of Healthcare services	Area of Region	Density of Healthcare Services (m ²)
150m	33	1,018.2	.0324
300m	68	4,729.05	.0143
450m	102	5,807.7	.0176
600m	114	7,465.5	.0153
750m	118	8,510.7	.0139

Table 2. The Proximity of Bus Stops to Primary Health Care Services in Chicago.

Distance from Health Care Services (m)	Number of Bus Stops	Area of Region	Density of Bus Stops (m ²)
150m	388	1,018.2	.3812
300m	1025	4,729.05	.2167
450m	1732	5,807.7	.2982
600m	2514	7,465.5	.3367
750m	3422	8,510.7	.4021

Table 3. The Proximity of Bus Stops to Hospitals in Chicago.

Distance from Hospital (m)	Number of Bus Stops	Area of Region	Density of Bus Stops (m ²)
150m	98	1,018.2	.0962
300m	347	4,729.05	.0734
450m	661	5,807.7	.1138
600m	1105	7,465.5	.1480
750m	1606	8,510.7	.1887

Table 4. Minimum, Maximum, and Average Distance of Healthcare Places to Bus Stops.

Statistics	# in meters
Average Distance of Healthcare Places from Bus Stops	90.91
Min. Distance	6
Max. Distance	890

Table 5. Minimum, Maximum, and Average Distance of Hospitals to Bus Stops.

Statistics	# in meters
Average distance of hospitals from bus stops	255.40
Min. Distance	27
Max. Distance	1203

Proximity of Health Care Services to Public Schools

The data revealed an adequate amount of primary health care services in Chicago. On the far Southeast side of Chicago, where the neighborhoods Roseland, Pullman, South Deering, and Riverdale are, the number of healthcare services given the amount of space is lower. The same is true for the far North side of Chicago, where the neighborhoods Norwood Park, Edison Park, and Jefferson Park are. When assessing the proximity to schools, most of the healthcare services are within the 450 meters range (seen in *Table 1*). The highest density of healthcare services was within the 150-meter range, which means that there are many health care locations closer to public schools, than farther away (see *Figure 1*).

From this, I deduced that the amount of health care locations within public schools is high, but transportation

from anywhere that is necessary, such as from being at home or work to a healthcare location, is also a major factor in receiving the proper health care. Therefore, the next results section addresses this issue. It would be useful to look at how much funding each school gets and the

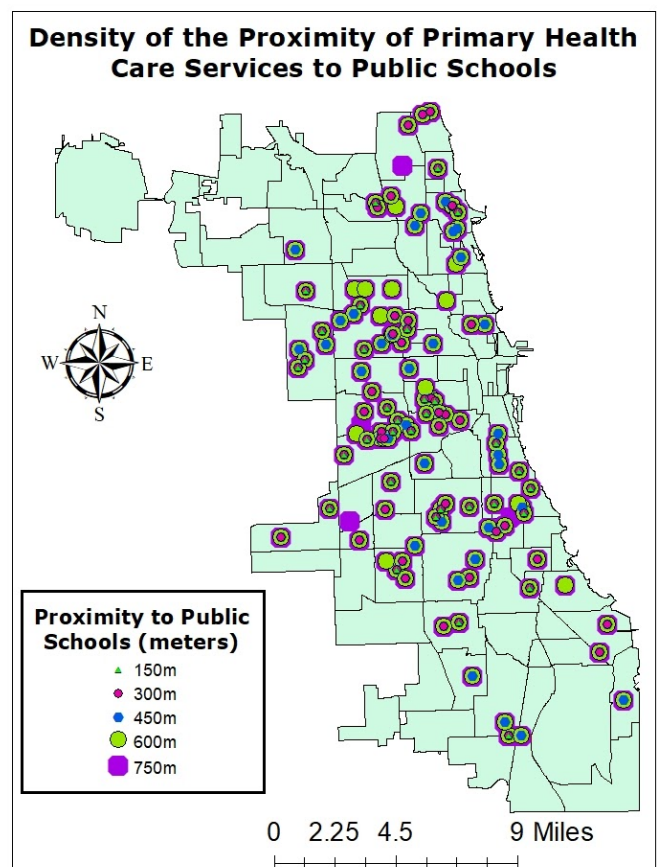


Figure 1. Density and Proximity of Primary Health Services in relation to Public Schools

neighborhood the public schools are in to see if there are discrepancies in the proximity to health care services within this data.

Proximity of Bus Stops to Health Care Services

Looking at the entire bus stop data, there are many bus stops in the Chicago area. Although, I assessed the bus stops close to primary health care service locations because people without cars may need to use public transportation, such as the bus, to receive the care they need. The social determinants of health have a major impact on a person's health, well-being and quality of life; one of these determinants is having access to adequate transportation systems. I found that the highest density of bus stops was within 750 meters of a primary health care location, at a density of .4021 meters squared. There are many bus stops that are within the 600 to 750 meter range from primary health care locations, meaning they are farther away. Although, the second highest density number found was within 150 meters of the primary health location (see *Table 2*). The range for the number of bus stops per health care service was 6 to 890 meters bus stops, with the average distance being 90.91 meters (see *Table 4*). Based on these results, their transportation access by bus routes to primary health service locations is adequate but can be improved. One question that arose from these results is to examine the neighborhoods by income and race, because Chicago is one of the most segregated cities in America, and the results could highlight any discrepancies present. I cross referenced data on the unemployment rates of community areas in Chicago, which was found and can be accessed on the Chicago city data portal at <https://data.cityofchicago.org/>. I thought it would be useful to see this data within the context of different community areas within Chicago because this can be used to assess transportation access by neighborhood.

Within Fuller park, the community area with the highest unemployment rate of 40%, the bus stops per health care place is on the lower end of the spectrum, within the 6-117 range of bus stops. There are two hospitals that are within a polygon and also within the neighborhood boundary of Fuller Park. The community area with the second highest unemployment rate is West Englewood, at 34.7%. It is within the 58-117 range of bus stops as seen on the map, and there is one primary health care location in the area, and three within the entire

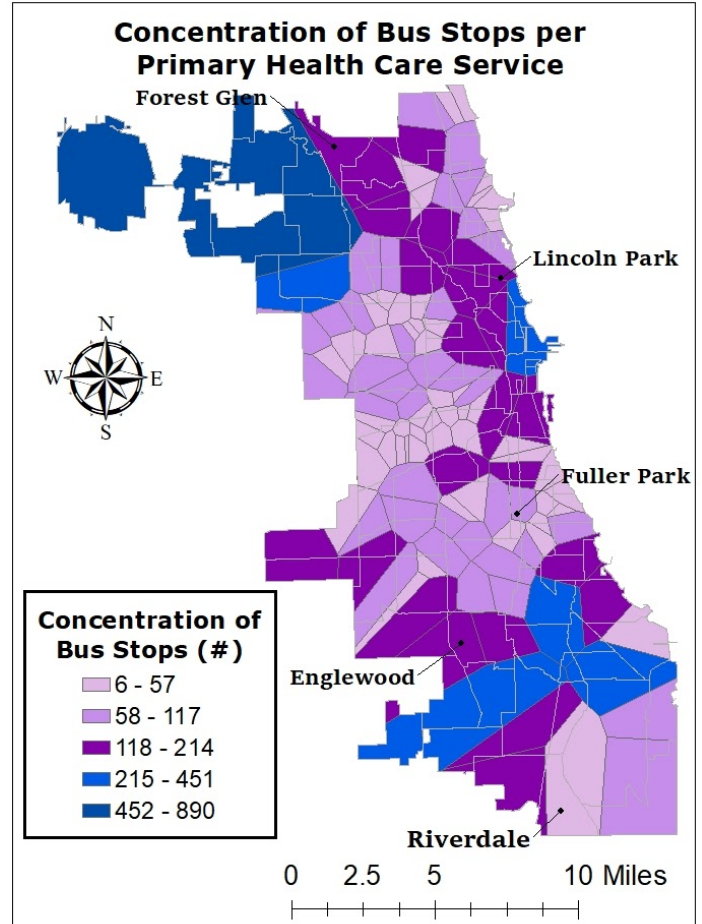


Figure 2. Concentration of bus stops with neighborhoods outlined.

Englewood county (see *Figure 2*). Other places worthy of noting that also have high unemployment rates are Washington Park (23.2), Burnside (23.4), Pullman (21), Riverdale (26.4), Auburn Gresham (24.2), and Grand Boulevard (20.6) (see *Table 6*). The community areas of interest were chosen because they have the lowest and highest unemployment rates in Chicago. Riverdale does not have a health care location within the area and is within the 6-57 range. On the other hand, Lincoln Park is one of the wealthiest community areas in Chicago and has a low unemployment rate of 4.5%. It is within the medium to high range of number of bus stops per health care location.

Community Area	Unemployment Rates (%)
Fuller Park	40
Grand Boulevard	20.6
Washington Park	23.2
Burnside	23.4
Pullman	21
Riverdale	26.40
Englewood	21.3
West Englewood	34.7
Auburn Gresham	24.2
Lincoln Park	4.5
Forest Glen	5.5
Lake View	4.7
Near South Side	5.7
Loop	4.2
O'Hare	4.7

Table 6. Unemployment rates (%) by Community Areas of Interest.

Forest Glen, another one of the richest counties to live in Chicago, has a low unemployment rate of 5.5% and is within the medium to high range of bus stops, of 118-451 (see *Figure 2*). There is a pattern of poorer counties with higher unemployment levels having less access to public transportation than the wealthier counties with lower unemployment levels.

Proximity of Bus stops to Hospitals

Data for hospitals, another health care service, was assessed in relation to the number of bus stops, and I found that the location of hospitals is mostly concentrated on the northern side of Chicago. The far Southwest side

of Chicago, including neighborhoods such as Englewood and Riverdale, have less hospitals near them and none in their respective counties. The distance of 750 meters from a hospital had the highest density of .1887 meters squared, and the second highest density was the 600-meter distance. This means that most bus stops are within a farther distance of 600 to 750 meters away from the nearest hospital (see *Table 3*). The range of distance of bus stops from hospitals is 27 to 1203 meters, with an average distance of 255.04 meters (see *Table 5*). This range is wider than the range for health care service locations, and there are more neighborhoods with a lack of hospitals nearby. The number of bus routes and access to public transportation can be improved with regards to hospitals. The data is also displayed in a map for a visual view (see *Figure 3*).

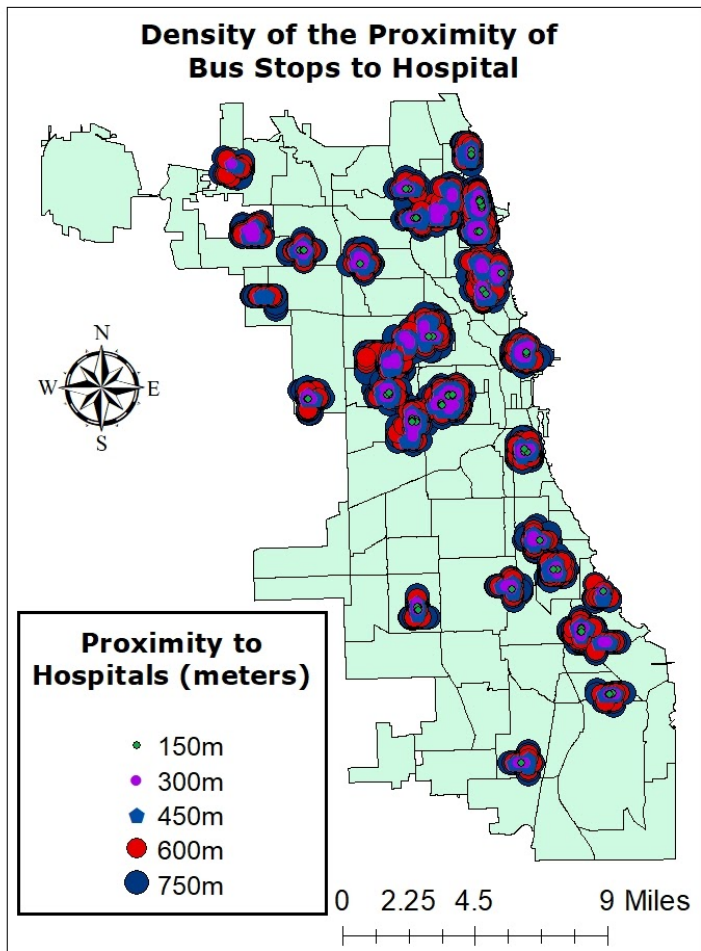


Figure 4. Density of the bus stops that are close to hospitals

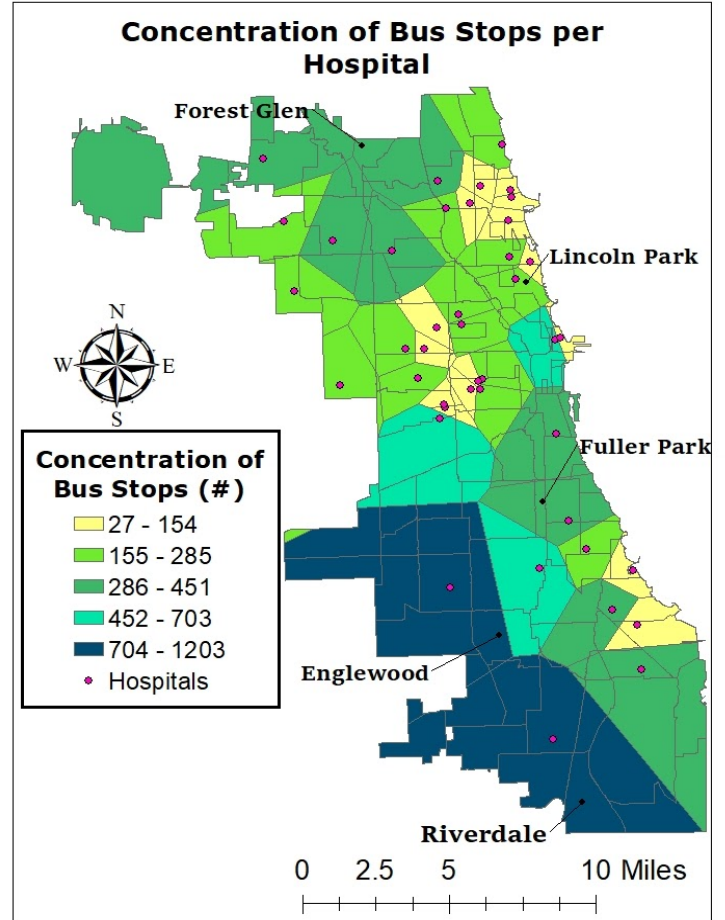


Figure 3. Displays the number of bus stops within each polygon, which represents a hospital.

The data on different community areas and unemployment rates were analyzed, and the results were not what I was predicting. Community areas such as Englewood, Fuller Park, and Riverdale had a higher number of bus stops within the area than areas like Forest Glen and Lincoln Park, which have lower rates of unemployment (see Figure 4). I was predicting the opposite. This could be explained because within community areas with higher rates of poverty, the demand for using public transportation can increase since people may be less likely to buy and own their own vehicle. Another explanation which makes sense to me is that, when looking at the map of the concentration of bus stops per hospital, the number of hospitals on the South side of Chicago is

significantly less than on the North side. The Thiessen polygons generated on the South side were larger than on the North and within many of the counties, there is not one hospital, especially on the far Southwest side (see *Figure 4*). This could be a reason why there are more bus stops because there are less resources available, like the number of hospitals in the area. It is important to have sufficient public transportation everywhere, but especially within areas with high concentrations of poverty, where taking public transportation is a necessity. This is only data on bus stops, and although useful, to continue this assessment I would look at other types of transportation like the metro subway system and also specifically data on the proximity to different neighborhoods. Another thing to assess is the funding for the hospitals depending on the neighborhood they are in; there may be a hospital in the area that one can take a bus to get to, but this begs the question of if the hospital is well funded and able to provide adequate care. This is another social determinant of health that can be analyzed in further detail.

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